Assignment 9: Complete MERN Deployment on Same EC2 Instance (Frontend + Backend)

Objective

Host both frontend and backend of MERN blog project on the same AWS EC2 instance using MongoDB Atlas.

---

1. Create MongoDB Atlas Cluster

Step 1: Login

- Open MongoDB Atlas
- Login / Create account

Step 2: Create Free Cluster

- Click **Create New Project** (if needed)
- Click **Build a Database**
- Choose:
- Free plan
- Provider: AWS
- Region: Mumbai (recommended)
- Cluster Name: Cluster0
- Disable sample dataset
- Enable automatic security setup
- Click **Create Deployment**

Wait 1–3 minutes.

Step 3: Create Database User

Go to: **Security → Database Access → Add New Database User**

Fill:

- Authentication Method: Password
- Username: `blogadmin`
- Password: `BlogApp123`

Privileges: Choose: `readWriteAnyDatabase`

Then click: **Add User**

Step 4: Network Access

Go to: **Security → Network Access / IP Access List**

Click: **ADD IP ADDRESS**

Choose: **Allow Access From Anywhere**

This adds:

```
0.0.0.0/0
```

Comment:

```
AWS EC2 Access
```

Wait until status becomes **Active**.

Step 5: Get MongoDB Connection String

Go to: **Clusters → Connect → Drivers**

Choose:

- Driver: Node.js
- Version: Latest

Copy connection string. Example:

```
mongodb+srv://blogadmin:BlogApp123@cluster0.xxxxx.mongodb.net/?
retryWrites=true&w=majority
```

Save this.

---

2. Create AWS EC2 Instance

Step 1: Launch Instance

AWS Console → EC2 → Launch Instance

Settings:

- Name: `mern-blog-server`
- AMI: Ubuntu Server 24.04 LTS
- Instance Type: t2.micro / t3.micro

Step 2: Key Pair

Create:

- Name: `mern-blog-key`
- Type: RSA
- Format: `.pem`

Download key.

Step 3: Security Group / Network Rules

Enable:

- SSH
- HTTP
- HTTPS

Add custom inbound rules:

Type	Port	Source
Custom TCP	5000	0.0.0.0/0
Custom TCP	5173	0.0.0.0/0

Launch instance.

Copy: **Public IPv4 address**

Example:

```
13.xx.xx.xx
```

---

3. Connect to EC2

Use browser EC2 terminal.

Run:

```
sudo apt update && sudo apt upgrade -y
sudo apt install git -y
```

---

4. Install Node.js

Run:

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
```

Verify:

```
node -v
npm -v
```

---

5. Clone Project

```
git clone https://github.com/devesh-bhangale-7789/mern-blog.git
cd mern-blog
```

---

6. Backend Setup

```
cd backend
npm install
```

Create env:

```
nano .env
```

Paste:

```
PORT=5000
MONGODB_URI=mongodb+srv://blogadmin:BlogApp123@cluster0.xxxxx.mongodb.net/?
retryWrites=true&w=majority
```

Save:

- CTRL + O
- Enter
- CTRL + X

Start backend:

```
npm run dev
```

Expected:

```
MongoDB connected
Server running on port 5000
```

Keep this terminal open.

---

7. Frontend Setup

Open NEW EC2 terminal.

```
cd ~/mern-blog/frontend
npm install
```

Create env:

```
nano .env
```

Paste:

```
VITE_API_URL=http://YOUR_PUBLIC_IP:5000/api
```

Example:

```
VITE_API_URL=http://13.xx.xx.xx:5000/api
```

Save.

Start frontend:

```
npm run dev -- --host
```

Expected:

```
VITE ready
Local: localhost:5173
Network: 172.xx.xx.xx:5173
```

Private IP shown here is normal.

---

8. Open Website

In your laptop browser:

```
http://YOUR_PUBLIC_IP:5173
```

Example:

```
http://13.xx.xx.xx:5173
```

DO NOT use:

```
172.xx.xx.xx
```

That is private EC2 IP.

---

9. If Refused to Connect

Check frontend started with host flag

Must use:

```
npm run dev -- --host
```

NOT:

```
npm run dev
```

Check security group

Inbound rules must include:

- 22
- 80
- 443
- 5000
- 5173

All source:

```
0.0.0.0/0
```

Check backend running

Backend terminal should show:

```
MongoDB connected
Server running on port 5000
```

Check firewall

Run:

```
sudo ufw status
```

If active:

```
sudo ufw allow 5000
sudo ufw allow 5173
```

---

10. Stop Everything

Frontend terminal:

```
CTRL + C
```

Backend terminal:

```
CTRL + C
```

AWS: Stop EC2 instance.

MongoDB: Delete cluster if assignment complete.